

What Helps When Telemetry Lies?

Four Pre-Registered Tests of Evidence Acquisition under Source Compromise

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Abstract—A threat hunter with a fixed budget must choose which telemetry sources to read, knowing that some of them may be controlled by the attacker. Many acquisition rules have been proposed for this setting, but they are rarely compared at equal spend, against attackers they were not designed for, with the comparison fixed before the data exist. We run four pre-registered experiments that each change one ingredient of a Bayesian acquisition policy while holding the shared posterior, the terminal decision, the horizon and the spend fixed. Three run on procedurally generated worlds (201,600, 499,200 and 537,600 policy-episodes) and one on a public industrial-control dataset. Two ingredients fail their gates: valuing integrity probes by their decision consequences adds nothing over hypothesis information under the same model (relative regret reduction 0.02, 95% CI $[-0.02, 0.06]$), and randomised or minimax acquisition does not beat a non-robust policy on a held-out attacker (-0.19 and -0.02). One ingredient passes every gate: scoring evidence by hypothesis information under a joint posterior over the hypothesis and every source’s compromise state cuts regret on a held-out attacker by 0.27 $[0.23, 0.30]$ relative to reliability-weighted information gain and by 0.22 $[0.19, 0.26]$ relative to joint-state information gain. The gain is large against cover-up and decoy attackers, small against misattribution, and absent against an attacker that adapts to the defender’s queries, where random acquisition is best. A replication on the HAI dataset was invalid by its own pre-registered check: the mapped evidence did not beat a decision made from the prior alone. The joint-belief result is therefore simulator-only.

Index Terms—threat hunting, adversarial telemetry, active hypothesis testing, value of information, pre-registration, negative results

I. INTRODUCTION

Attackers tamper with the evidence defenders rely on. Log deletion, forwarder sabotage and sensor spoofing are catalogued techniques (MITRE ATT&CK T1562, “Impair Defenses” [1]), and stale-value spoofing of process variables is a standard attack on industrial control systems [2]. A hunter who investigates an alert under a budget therefore faces two coupled questions: which source to read next, and how far to believe it.

Each piece of this problem has a long history. Choosing the next measurement by expected entropy reduction over a joint fault model goes back to the General Diagnostic Engine [3]; decision-theoretic troubleshooting values tests by their effect on the repair decision [4]; active sequential hypothesis testing gives asymptotically optimal sensing rules [5]; and Byzantine detection studies inference when some sensors lie [6], [7]. Recent work combines a belief over compromised sensors with

active probing in cyber-physical control [8] and learns cost-aware sensor scheduling with a Byzantine detector [9].

What is missing is a controlled answer to a practical question: *which ingredient of such a policy actually reduces decision errors when sources lie?* A new policy usually differs from its baseline in several ways at once (belief model, objective, look-ahead, cost handling), is evaluated against the attacker its model assumes, and is compared at unequal spend. Any of these can manufacture an apparent gain.

We address this with an ablation discipline rather than a new method. Every policy in this paper shares the same exact posterior, the same Bayes terminal decision and exactly the same spend; policies differ only in how they score the next action. Each experiment changes one ingredient, names its comparator, fixes a practical threshold (a 10% relative cut in regret) and its kill gates in a committed protocol, and is tested on an attacker written after the policies were frozen.

Contributions.

- 1) A spend-matched, pre-registered testbed for evidence acquisition under source compromise: procedurally generated worlds, six attacker families of which two were held out at design time, and frozen gate code (Sec. IV).
- 2) Two pre-registered negative results. Valuing source-integrity probes by their decision consequences adds no robust gain over hypothesis information under the same joint model, and randomised or minimax acquisition loses to the non-robust comparator on a held-out attacker (Sec. V-A).
- 3) One scoped positive result. At equal objective, horizon and cost handling, replacing a reliability-weighted belief with a joint belief over hypothesis and source states reduces regret by 0.27 on a held-out attacker; we report where the gain vanishes (Sec. V-B).
- 4) A real-data attempt on the HAI industrial-control dataset that failed its own validity check, with the reasons, so that the positive result is reported as simulator-only (Sec. VI).

We do not claim the joint belief model as new; it is classical [3], [4], [7]. The contribution is the controlled evidence about which ingredient matters, and where it stops mattering.

II. RELATED WORK

Diagnosis and troubleshooting. GDE selects measurements by minimum expected entropy over component states

[3]; Bayesian-network test selection does the same at scale [10]; decision-theoretic troubleshooting values tests by their effect on the repair decision [4]. Faults there are not strategic, and observations of healthy components are trusted. Our joint-state objective J is the GDE criterion and our decision-value probe P follows the troubleshooting criterion.

Sensing with adversarial sources. Byzantine detection studies fusion when a fraction of sensors lie, mostly through error exponents and game equilibria with all sensors observed [6]. Soltanmohammadi et al. infer node identity jointly with the hypothesis by EM [7], the closest inference-side match to our belief model, but without a choice of which source to read. Andersson and Dán maintain a Bayesian network over sensor attack states and choose a probing input that maximises a KL divergence, with the goal of control recovery [8]. Zhong et al. learn sensor scheduling with probing costs and a separate Byzantine detector [9]. MATE estimates per-agent trust as a hidden state for fusion [11]. None of these ablates the belief model against the objective at equal spend, or evaluates on attackers written after the policy.

Active learning with unreliable oracles. Proactive learning selects among imperfect, cost-varying oracles [12]; cost-aware active learning divides information by cost [13], [14]. Oracle noise there is not strategic. Recent work attacks the acquisition step itself [15], [16]; it does not study which acquisition rule resists such attacks. Robust Bayesian experimental design optimises the worst case over model misspecification [17], which motivates our minimax candidate.

Security operations. POMDP formulations plan sensing and defence under noisy alerts [18], [19] but treat sensor noise as non-strategic. Provenance triage [20], tamper-evident logging [21] and hunting under anti-forensics [22] motivate our integrity probe and our compromised-source model, but do not choose evidence decision-theoretically.

III. PROBLEM AND POLICIES

State. A hidden hypothesis $H \in \{\text{benign}, \text{intrusion}_1, \text{intrusion}_2\}$ and, for each of K sources, a state $S_q \in \{\text{healthy}, \text{degraded}, \text{compromised}\}$. A *query* of source q returns an outcome o indicating a hypothesis; a healthy source reports correctly with probability acc_q , a degraded one uniformly, and a compromised one (in the defender’s model) returns a forged outcome with probability 0.9: a cover-up (“benign”) under an intrusion, a false flag under benign. A *probe* of q is an integrity check (canary, heartbeat or cross-sensor check) that returns PASS or FAIL with state-dependent rates. Compromise is less likely under benign activity: $P(S_q=C \mid \text{benign}) = \kappa P(S_q=C \mid \text{intrusion})$.

Decision and regret. After the budget is spent, the defender takes $D^* = \arg \max_D \mathbb{E}[U(D, H) \mid E]$ over $\{\text{dismiss}, \text{contain}, \text{block}\}$: 0 if correct, $-L_{\text{miss}}$ for dismissing an intrusion, $-0.5L_{\text{miss}}$ for the wrong containment, -1 for a false alarm. Regret is $U(D^*(H), H) - U(D, H)$ divided by L_{miss} .

Shared machinery. Every policy sees the same exact posterior $P(H, S_1, \dots, S_K \mid E)$, and the simulator, not the policy, applies the same Bayes decision. Costs are integers

with a fixed budget, and every policy must act while it can afford an action, so every policy spends exactly the budget in every episode (checked in every run). Policies never see the true hypothesis, source states or attacker identity; leakage tests enforce this.

Policies differ only in scoring. A score has three parts: the *model* used for look-ahead, the *objective* credited per outcome, and the *horizon*. With horizon 2 and cost-awareness, $\text{score}(a) = \max(V_1(a)/c_a, \max_b V_2(a, b)/(c_a + c_b))$. The four policies that matter here are:

- R (P3_RHAT_COST_LA2): reliability-weighted model. Look-ahead treats a source as $\hat{r}_q p_{\text{nom}} + (1 - \hat{r}_q)$ uniform, with $\hat{r}_q$ the current posterior probability that q is healthy, and H and S independent. Objective: entropy reduction of H .
- C (P_JOINT_HEIG_COST): exact joint model; objective: entropy reduction of H .
- J (P_JOINT_EIG_COST): exact joint model; objective: entropy reduction of (H, S) , the GDE criterion.
- P (P_PROBE): exact joint model; objective: expected gain in the Bayes-decision utility (value of information), falling back to H -information on VoI plateaus.

All four use horizon 2 and divide by cost. C and R differ only in the belief model; C and J only in the objective. References include random acquisition, nominal EIG that ignores cost and compromise (P3_NOM), and, in E2, two robust variants of J .

IV. EXPERIMENTAL SETUP

Worlds. Each unit of analysis is a world drawn from a frozen distribution: $K \sim U\{5..8\}$ sources; source accuracy $U(0.45, 0.95)$; query cost $U\{1..4\}$; probe-cost ratio $\log\text{-}U(0.25, 2)$; compromise prior $U(0.02, 0.40)$ with per-source jitter; benign coupling $\kappa \sim U(0, 1)$; probe detection $U(0.5, 0.99)$ and false-fail $U(0, 0.1)$; loss ratio $L_{\text{miss}} \log\text{-}U(1, 20)$; budget 12. No world is hand-built.

Attackers. An attacker controls $B \in \{1, 2, 3\}$ sources and is isolated from the policies. Six families: *cover-up* (the design family, matching the defender’s model); *misattribution* (targets the most trusted sources and pushes the wrong containment); *trust-harvest* (adaptive: honest until queried, then takes the source over, lies low after a probe); *decoy-migrate* (held out in E2: a sticky decoy class, migrates when probed); *collude-cheap* (held out in E3: controls the cheapest sources, which all tell the same lie). Within a world every policy faces the same truths and random numbers (common random numbers), 12 episodes per (world, condition).

Endpoint and statistics. For candidate X and comparator Y , $\bar{y}_w$ is a world’s mean normalised regret over the primary cells and $\text{RRR} = 1 - \sum_w \bar{y}_w(X) / \sum_w \bar{y}_w(Y)$. A gate passes only if $\text{RRR} \geq 0.10$ and the one-sided paired-bootstrap lower bound (10,000 resamples, fixed seed) exceeds 0. The number of worlds came from a blinded calibration on separate seeds that output only the comparator’s mean and the spread of the difference. E1’s calibration asked for 140 worlds; E2 and E3 asked for more than the pre-registered cap of 400 and ran 400.

TABLE I
THE FOUR PRE-REGISTERED EXPERIMENTS. EACH CHANGES ONE
INGREDIENT AGAINST NAMED COMPARATORS.

Ingredient tested	Comparators	Units
E1 decision-valued probing (P)	J ; R , C	140 worlds
E2 randomised / minimax acquisition	J ; random	400 worlds
E3 joint belief model (C)	R and J	400 worlds
E4 E3 on real data (HAI)	R and J	100 episodes

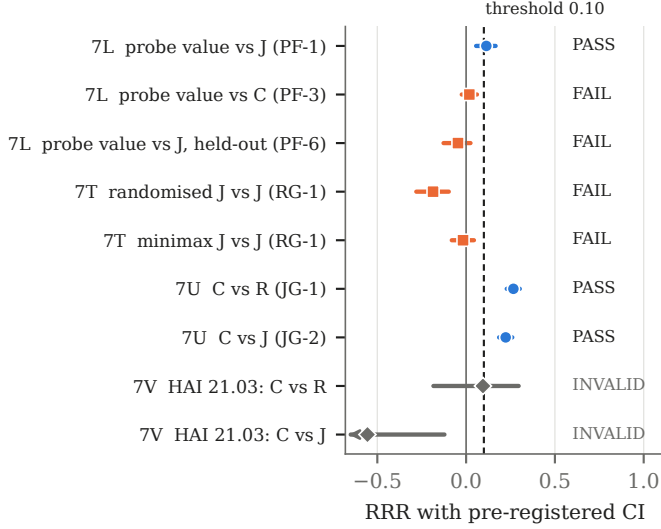


Fig. 1. Primary contrasts of all four experiments: RRR (marker) and pre-registered CI (bar) against the 0.10 threshold. “PASS”/“FAIL” is the gate on that row; E1 passed its headline gate but failed three others and was killed. E4 rows are shown for completeness; the run was invalid. Bars below -0.65 are truncated (arrow).

Each experiment also has a validity check: the attacker must raise the regret of nominal EIG, otherwise the run is *invalid*.

Pre-registration. For each experiment the protocol, gates, configuration, gate code and tests were committed before any calibration or main run; the runners refuse a dirty tree, a changed configuration or an existing output, and the analysis scripts refuse to overwrite a verdict. Table I lists the four experiments.

V. RESULTS

Fig. 1 and Table II show every primary contrast with its pre-registered interval. Validity checks passed in E1–E3 and failed in E4; spend matching and leakage gates passed in all four.

A. Negative results

E1: decision-valued probing. P beat J on the pooled known attackers (RRR 0.11 [0.06, 0.17]), but the experiment was killed by three gates. Against C , which uses the same joint model with plain H -information, the gain vanished (0.02 [-0.02, 0.06]). On the held-out misattribution attacker it was negative (-0.05 [-0.13, 0.02]). The pooled gain came from the design attacker, whose behaviour matches the defender’s compromised-source model (0.49 [0.41, 0.56]). Probing was

TABLE II
PRIMARY CONTRASTS. RRR [CI]: 95% FOR E1, E3, E4; 97.5% FOR E2
(TWO CANDIDATES, BONFERRONI). P : DECISION-VALUE PROBE; C , R , J
AS IN SEC. III.

Contrast	Gate	RRR [CI]	Result
7L P vs J , known families	PF-1	0.114 [0.056, 0.166]	pass
P vs C , known families	PF-3	0.018 [-0.025, 0.061]	fail
P vs J , held-out misattrib.	PF-6	-0.046 [-0.127, 0.025]	fail
7T Rand. J vs J , held-out	RG-1	-0.186 [-0.281, -0.097]	fail
Minimax J vs J , held-out	RG-1	-0.017 [-0.081, 0.044]	fail
7U C vs R , held-out	JG-1	0.266 [0.229, 0.303]	pass
C vs J , held-out	JG-2	0.223 [0.186, 0.260]	pass
7V C vs R , HAI 21.03	HG-1	0.095 [-0.184, 0.296]	n/a
C vs J , HAI 21.03	HG-2	-0.556 [-1.438, -0.122]	n/a

also untargeted: only about a third of P ’s probes went to sources the posterior considered suspicious, against a pre-registered floor of one half.

E2: robust acquisition. E1’s exploratory data suggested that model-based policies fail when the attacker does not match their model. E2 tested two principled fixes, each changing J in one way: sampling actions in proportion to their score, and scoring by the worst case over three compromised-source models. On a new held-out attacker neither beat J : randomisation lost clearly (-0.19 [-0.28, -0.10]) and the minimax policy tied (-0.02 [-0.08, 0.04]) while losing on the known families.

B. Scoped positive result

Origin, disclosed. In E1 and E2, C had markedly lower regret than both R and J in exploratory tables. E3 tested this prospectively, on new world seeds and a new held-out attacker (collude-cheap) written before any E3 run. No policy, threshold or world parameter was changed.

E3: joint belief. Every gate passed. On collude-cheap, C reduced regret by 0.27 [0.23, 0.30] relative to R and by 0.22 [0.19, 0.26] relative to J , with a positive RRR in each attacker budget and in both halves of the worlds, and lower regret than R on clean worlds. Because C and R share the objective, horizon, cost handling, posterior, decision and spend, the gain is attributable to the look-ahead belief model: modelling that a compromised source reports a cover-up, and that its state persists and is correlated with H , rather than treating it as symmetric noise independent of H .

Where it stops. Fig. 2 and Table III show the limits, which travel with the claim. (i) Against misattribution the gain over R is small (0.05 [0.01, 0.10]). (ii) Against the adaptive trust-harvest attacker there is no gain over J (-0.03 [-0.09, 0.03]), and random acquisition has the lowest regret of all policies; the same held in E1 and E2. (iii) C is not the best policy on collude-cheap: nominal EIG, which ignores cost, avoids the cheap sources the attacker controls and does better. The pre-registered claim is a comparison among cost-aware policies, not “best against this attacker”.

VI. REAL-DATA ATTEMPT: HAI

Mapping. HAI is a public hardware-in-the-loop industrial-control dataset with labelled attacks [2], [23]; several of its

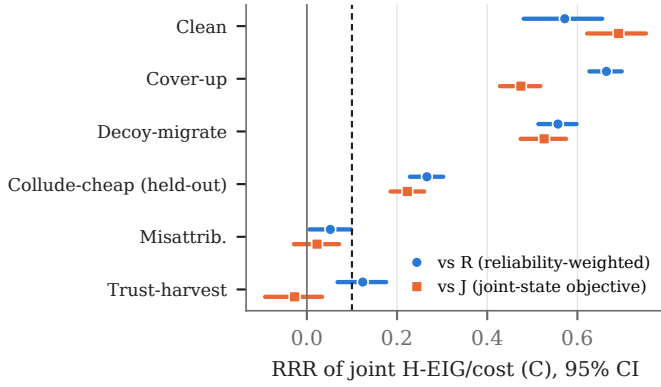


Fig. 2. E3 scope map (descriptive, not gated): RRR of C against R (circles) and J (squares) by attacker family, 95% CI. The gain is large against cover-up, decoy and colluding attackers, small against misattribution, and absent against trust-harvest relative to J .

TABLE III

E3 MEAN NORMALISED REGRET BY ATTACKER FAMILY (400 WORLDS; 537,600 POLICY-EPISODES IN TOTAL). BOLD: LOWEST IN THE COLUMN AMONG ALL SEVEN POLICIES.

Policy	Clean	Cover	Decoy	Collude	Misattr.	Harvest
Random	0.100	0.183	0.156	0.170	0.185	0.126
EIG (nominal)	0.051	0.319	0.137	0.132	0.280	0.259
EIG- $\hat{r}$ /cost LA2 (R)	0.033	0.229	0.122	0.212	0.190	0.254
Joint (H,S)-EIG/cost (J)	0.046	0.146	0.114	0.200	0.184	0.217
Joint H-EIG/cost (C)	0.014	0.077	0.054	0.156	0.180	0.223
Decision-value probe	0.017	0.076	0.063	0.143	0.195	0.205

scenarios hold a process variable at a stale value while the process moves, which is a real cover-up. We used HAI 20.07 for development (detectors and model fitting, 76 episodes), HAI 21.03 for the verdict and HAI 22.04 as secondary. An episode is one attack segment or a sampled normal window; hypotheses are normal, attack confined to the boiler process, or attack touching the other processes. Nine process variables and controller outputs are sources. A query reads a source's change alarm over a 30-second window (threshold at the 99th percentile of attack-free training data); a probe checks the source against a ridge regression on the other points of its process. The k -th read of a source returns the k -th window, up to six reads. The defender model was fitted on development episodes only and applied unchanged; the protocol was committed before any confirmatory episode was built, and an in-sample look at the development data was disclosed in it.

Outcome: invalid. The validity check requires nominal EIG to beat a decision taken from the prior alone. It did not: on 21.03 the difference was 0.040 $[-0.041, 0.119]$ (100 episodes, 50 attacks, 19 with spoofing), and on 22.04 every evidence-reading policy did worse than the prior ($-0.083 [-0.172, 0.004]$). The comparison of belief models is therefore not tested. Table IV shows why: missed boiler attacks dominate regret, no deterministic policy ever probed (the development probe failed 0.355 of healthy and 0.674 of spoofed readings,

TABLE IV
E4 (HAI) MEAN NORMALISED REGRET, ALL EPISODES. THE VALIDITY CHECK COMPARES NOMINAL EIG WITH THE PRIOR-ONLY DECISION.

Policy	HAI 21.03	HAI 22.04
Prior only (no evidence)	0.215	0.248
Random	0.223	0.261
EIG (nominal)	0.175	0.331
EIG- $\hat{r}$ /cost LA2 (R)	0.232	0.363
Joint (H,S)-EIG/cost (J)	0.135	0.348
Joint H-EIG/cost (C)	0.210	0.358
Decision-value probe	0.105	0.297

at twice a query's cost), and the evidence is weak.

An exploratory warning. Although not interpretable under the protocol, on 21.03 J had lower regret than C ($-0.56 [-1.44, -0.12]$); on 22.04 the two were level ($-0.03 [-0.20, 0.11]$). If a future replication passes the validity check, the E3 result could fail there. We do not re-run E4; a new attempt needs a new protocol and a detector that passes the validity check first.

VII. THREATS TO VALIDITY

External validity. E1–E3 use one simulator family and one world distribution. The only real-data attempt was invalid, and its exploratory direction on 21.03 disagreed with E3. The positive result must be read as simulator-only.

Model match. The defender's compromised-source model is the cover-up model. Attackers that lie in that direction (cover-up, decoy, collude) are where C gains most; misattribution and adaptive attackers are where it does not. Held-out attackers reduce, but do not remove, this bias: the same researcher wrote the policies and the attackers.

Exploratory origin. E3's hypothesis came from E1 and E2 data after their verdicts. E3 is its first prospective test, with new seeds and a new attacker; a second independent test has not been done.

Construct validity. Regret is normalised per world and RRR is a ratio of sums, so worlds with large regret weigh more. Results are reported per budget and per world half to show the effect is not driven by a few worlds. The HAI mapping (change alarms, a regression probe, three coarse hypotheses) is one of many possible and may be too weak.

Multiplicity. Each experiment controls its own error rate (intersection–union across comparators; Bonferroni across E2's two candidates). Across the four experiments we report all outcomes, including the failures.

VIII. CONCLUSION

When sources may lie, what helped in our experiments was not a cleverer objective or a robustness wrapper but a better look-ahead model: one that represents each source's compromise state jointly with the hypothesis. That result is scoped to non-adaptive attackers whose lies resemble the defender's model, it vanishes against an attacker that exploits the defender's choices (where random acquisition is best), and it is not yet supported by real data. Decision-valued probing

and robust acquisition both failed pre-registered tests. We hope the spend-matched, held-out-attacker protocol is useful beyond these results: in this problem, three of four ideas that looked good on the data that suggested them did not survive a test fixed in advance.

ARTIFACT AVAILABILITY

Code, protocols, gate code, raw outputs, statistics and verdicts are in the public repository <https://github.com/ftaakash/Hades> (branch `claude/hades2-probe-pilot-gemc0j`).

Experiment IDs: E1 `phase7l_probe_pilot_v2`, E2 `phase7t_robust_pilot_v1`, E3 `phase7u_joint_belief_v1`, E4 `phase7v_hai_replication_v1`. Every number in this paper is regenerated from the raw outputs by `scripts/paper/recompute_headlines.py` and `scripts/paper/make_figures.py`. HAI is available from <https://github.com/icsdataset/hai> under CC BY-SA 4.0.

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